

Lie Theory for the Impatient

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For my dearest dear, Hui.

Abstract

This document collects a variety of definitions and results from Lie theory. To avoid distraction, all vector spaces are over $\mathbb{C}$, and all manifolds are real.

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1 Lie Algebras

Before starting, let's overview where we are going.

- Section 1.1 provides the basic definitions of the study of Lie algebras, such as Lie algebras and their representations.
- Section 1.2 provides some facts about an “easy” subclass of Lie algebras. In short, nilpotent and solvable Lie algebras are those which are amenable to study by inductive arguments.
- Section 1.3 begins the study of the remaining Lie algebras, which are the semisimple ones.
- Sections 1.4 to 1.6 classifies semisimple Lie algebras.
- Sections 1.7 and 1.8 study the representation theory of semisimple Lie algebras.

1.1 Starting Out

Definition 1 (Lie algebra). A *Lie algebra* is a vector space $\mathfrak{g}$ equipped with a *Lie bracket* $[-, -]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying the following properties.

- (i) Bilinear: the map $[-, -]$ is bilinear.
- (ii) Alternating: $[X, X] = 0$ for all $X \in \mathfrak{g}$.
- (iii) Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ for all $X, Y, Z \in \mathfrak{g}$.

A morphism of Lie algebras is a linear map preserving the bracket.

Example 2. For any (possibly non-commutative, non-unital) algebra A over $\mathbb{C}$, one can define the Lie bracket $[-, -]: A \times A \rightarrow A$ by

$$[a, b] := ab - ba.$$

For example, applying this to $A = \text{End}(V)$ where V is a vector space gives the Lie algebra $\mathfrak{gl}(V)$. If further $V = \mathbb{C}^n$, we abbreviate $\mathfrak{gl}(V)$ to $\mathfrak{gl}_n(\mathbb{C})$.

Example 3. The subspace

$$\mathfrak{sl}_n(\mathbb{C}) := \{X \in \mathfrak{gl}_n(\mathbb{C}) : \text{tr } X = 0\}$$

is a Lie subalgebra.

Example 4. The subspace of alternating matrices is a Lie subalgebra, called $\mathfrak{o}_n(\mathbb{C}) \subseteq \mathfrak{gl}_n(\mathbb{C})$.

Definition 5 (representation). A *representation* of a Lie algebra $\mathfrak{g}$ is a vector space V along with a morphism $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$. A morphism of representations is a morphism of the underlying vector spaces which preserves the $\mathfrak{g}$ -action.

Example 6 (adjoint). The adjoint representation of a Lie algebra $\mathfrak{g}$ is the morphism $\text{ad}_\bullet: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ defined by

$$\text{ad}_X(Y) := [X, Y].$$

The fact that $\text{ad}_\bullet$ preserves the Lie bracket is equivalent to the Jacobi identity.

Example 7. For any representation V of $\mathfrak{g}$, there is a dual representation $V^\vee$ whose underlying vector space is the dual, and $X \in \mathfrak{g}$ acts on $\ell: V \rightarrow \mathbb{C}$ by $(X\ell)(v) := -\ell(Xv)$.

Example 8. One can take sums of representations. For two representations V and W , one can also take a tensor product: $X \in \mathfrak{g}$ acts by $X(v \otimes w) = Xv \otimes w + v \otimes Xw$.

Definition 9 (irreducible). A representation $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is *irreducible* if and only if $V \neq 0$, and V admits no nonzero proper subrepresentations. A representation is *completely reducible* if and only if it is a direct sum of irreducible representations.

Example 10. Let $\mathfrak{b} \subseteq \mathfrak{gl}_2(\mathbb{C})$ be the Lie subalgebra of upper-triangular matrices. Then the representation $\mathfrak{b} \rightarrow \mathfrak{gl}_2(\mathbb{C})$ is not completely reducible.

Lemma 11 (Schur). Let $\rho_V: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ and $\rho_W: \mathfrak{g} \rightarrow \mathfrak{gl}(W)$ be representations, and let $\varphi: V \rightarrow W$ be a nonzero morphism of representations.

- (a) If V is irreducible, then φ is injective.
- (b) If W is irreducible, then φ is surjective.
- (c) If both V and W are irreducible, then φ is an isomorphism.

Sketch. Here, (c) follows from (a) and (b). We only sketch (a): because φ is nonzero, $\ker \varphi$ is a proper subrepresentation of V , so $\ker \varphi = 0$ because V is irreducible. ■

Example 12. Let the natural action of $\mathfrak{sl}_2(\mathbb{C})$ on $\mathbb{C}^2$ be irreducible. It turns out that the irreducible representations of $\mathfrak{sl}_2(\mathbb{C})$ are exactly the symmetric powers of $\mathbb{C}^2$. To see this, show that $\text{diag}(+1, -1)$ acts semisimply on any irreducible representation, and then one can parameterize the irreducible representations by the highest eigenvalue of $\text{diag}(+1, -1)$.

Definition 13 (universal enveloping algebra). Fix a Lie algebra $\mathfrak{g}$. Then the *universal enveloping algebra* $U\mathfrak{g}$ of $\mathfrak{g}$ is the free complex algebra generated by $\mathfrak{g}$ along with the relations

$$XY - YX = [X, Y]$$

for each $X, Y \in \mathfrak{g}$.

Remark 14. The data of a representation $\mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is equivalent to the data of a ring homomorphism $U\mathfrak{g} \rightarrow \text{End}(V)$.

Theorem 15 (Poincaré–Birkhoff–Witt). Fix a Lie algebra $\mathfrak{g}$, and let $\{X_i\}_{i=1}^n$ be an (ordered) basis. Then the monomials

$$\{X_{i_1} \cdots X_{i_m} : i_1 \leq i_2 \leq \cdots \leq i_m\}$$

form a basis of $U\mathfrak{g}$.

Sketch. Certainly $U\mathfrak{g}$ is spanned by monomials, and one can use the relations $XY = YX + [X, Y]$ in order to rearrange the entries of the monomial. Thus, the given monomials span. Checking that they are linearly independent amounts to a long and technical argument. ■

1.2 Nilpotent and Solvable Lie Algebras

Definition 16 (ideal). Fix a Lie algebra $\mathfrak{g}$. Then an *ideal* $I \subseteq \mathfrak{g}$ is a subspace such that $[X, I] \subseteq I$ for any $X \in \mathfrak{g}$. Equivalently, I is a $\mathfrak{g}$ -subrepresentation of $\mathfrak{g}$.

Example 17 (center). Fix a Lie algebra $\mathfrak{g}$. Then the *center* is the subspace

$$\mathfrak{z}(\mathfrak{g}) := \{X \in \mathfrak{g} : [X, \mathfrak{g}] = 0\}.$$

The center is an ideal. It is kernel the adjoint representation $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$.

Example 18 (commutator). Fix a Lie algebra $\mathfrak{g}$. Then the *commutator* is the subspace $[\mathfrak{g}, \mathfrak{g}]$. By the Jacobi identity, $[\mathfrak{g}, \mathfrak{g}]$ is a Lie ideal.

Remark 19. If I is an ideal of a Lie algebra $\mathfrak{g}$, the quotient space $\mathfrak{g}/I$ is a Lie algebra, and there is a short exact sequence

$$0 \rightarrow I \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}/I \rightarrow 0$$

of $\mathfrak{g}$ -modules.

Definition 20. A Lie algebra $\mathfrak{g}$ is *abelian* if and only if $[\mathfrak{g}, \mathfrak{g}] = 0$.

Example 21. For any Lie algebra, the quotient $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian.

Definition 22 (nilpotent). A Lie algebra $\mathfrak{g}$ is *nilpotent* if and only if there is a sequence

$$\mathfrak{g} = \mathfrak{a}_0 \supseteq \mathfrak{a}_1 \supseteq \cdots \supseteq \mathfrak{a}_k = 0$$

such that $\mathfrak{a}_{i+1}$ is an ideal of $\mathfrak{a}_i$ with abelian quotient.

Remark 23. It is equivalent to check if the sequence $\mathfrak{a}_{i+1} := [\mathfrak{g}, \mathfrak{a}_i]$ eventually stabilizes at 0.

Example 24. The subspace of strictly upper-triangular matrices of $\mathfrak{gl}_n(\mathbb{C})$ is nilpotent.

Definition 25 (solvable). A Lie algebra $\mathfrak{g}$ is *solvable* if and only if there is a sequence

$$\mathfrak{g} = \mathfrak{a}_0 \supseteq \mathfrak{a}_1 \supseteq \cdots \supseteq \mathfrak{a}_k = 0$$

such that each $\mathfrak{a}_i$ is an ideal of $\mathfrak{g}$ such that $\mathfrak{g}/\mathfrak{a}_i$ is abelian.

Remark 26. It is equivalent to check if the sequence $\mathfrak{a}_i := [\mathfrak{a}_{i+1}, \mathfrak{a}_{i+1}]$ eventually stabilizes at 0.

Example 27. Any nilpotent Lie algebra is solvable.

Example 28. The subspace of upper-triangular matrices of $\mathfrak{gl}_n(\mathbb{C})$ is solvable but not nilpotent.

Theorem 29 (Lie). Let $\mathfrak{g}$ be a solvable Lie algebra. For any representation V of $\mathfrak{g}$, there is a basis in which $\mathfrak{g}$ acts by upper-triangular matrices.

Sketch. By induction on $\dim V$, it is enough to show that the action of $\mathfrak{g}$ on V admits a nonzero eigenvector. For this, we induct on $\dim \mathfrak{g}$. By solvability, one may find a codimension-1 ideal $\mathfrak{g}' \subseteq \mathfrak{g}$; choose any $X \in \mathfrak{g} \setminus \mathfrak{g}'$. Now, induction tells us that V admits a nonzero eigenvector v for $\mathfrak{g}'$. With some difficulty, can show that

$$W := \text{span}_{\mathbb{C}} X^{\mathbb{N}} v$$

is a $\mathfrak{g}$ -subrepresentation of V , and it turns out that $\mathfrak{g}'$ acts diagonally on all of W . Thus, any eigenvector of X acting on W will do. ■

Corollary 30. Let $\mathfrak{g}$ be a Lie algebra. Then $\mathfrak{g}$ is solvable if and only if $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.

Sketch. If $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent, then one can show that $\mathfrak{g}$ is solvable by building ideals from the short exact sequence

$$0 \rightarrow [\mathfrak{g}, \mathfrak{g}] \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \rightarrow 0.$$

Conversely, if $\mathfrak{g}$ is solvable, then by Theorem 29, one can find a basis of $\mathfrak{g}$ on which the adjoint action of $\mathfrak{g}$ on $\mathfrak{g}$ is given by upper-triangular matrices. Thus, the action of $[\mathfrak{g}, \mathfrak{g}]$ is given by strictly upper-triangular matrices. This is enough to show $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent. ■

Remark 31. Similarly, if $\mathfrak{g}$ is nilpotent, one can show that any representation V admits a basis on which $\mathfrak{g}$ acts by strictly upper-triangular matrices.

Theorem 32 (Engel). Let $\mathfrak{g}$ be a Lie algebra. Then $\mathfrak{g}$ is nilpotent if and only if the operator $\text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ is nilpotent for all $X \in \mathfrak{g}$.

Sketch. If ad_X is nilpotent for all $X \in \mathfrak{g}$, then $[\mathfrak{g}, [\mathfrak{g}, [\mathfrak{g}, \dots, [\mathfrak{g}, \mathfrak{g}]]]] = 0$ for a high enough nesting level, so $\mathfrak{g}$ is nilpotent. Conversely, if $\mathfrak{g}$ is nilpotent, then Remark 31 grants a basis of $\mathfrak{g}$ upon which $\mathfrak{g}$ acts by strictly upper-triangular matrices, so ad_X is nilpotent for all $X \in \mathfrak{g}$. ■

1.3 Simple and Semisimple Lie Algebras

Definition 33 (simple). A Lie algebra $\mathfrak{g}$ is *simple* if and only if it is not abelian and admits no nonzero proper ideals.

Example 34. One can check by hand that $\mathfrak{sl}_2(\mathbb{C})$ is a simple Lie algebra.

Definition 35 (semisimple). A Lie algebra $\mathfrak{g}$ is *semisimple* if and only if it admits no nonzero solvable ideals.

Example 36. Any simple Lie algebra is semisimple. We will have many more examples shortly.

Remark 37. It turns out that any semisimple Lie algebra $\mathfrak{g}$ is a direct sum of simple ones. In short, one can decompose the adjoint action of $\mathfrak{g}$ on $\mathfrak{g}$ into a sum of irreducible subrepresentations, and each irreducible subrepresentation is a simple Lie subalgebra.

Remark 38. If $\mathfrak{g}$ is semisimple, then $\mathfrak{z}(\mathfrak{g}) = 0$, so the adjoint representation $\mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is injective.

Definition 39 (reductive). A Lie algebra $\mathfrak{g}$ is *reductive* if and only if $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ is semisimple.

Example 40. Because $\mathfrak{z}(\mathfrak{gl}_2(\mathbb{C}))$ consists of the scalar matrices, the Lie algebra $\mathfrak{gl}_2(\mathbb{C})$ is reductive.

Remark 41. A reductive Lie algebra $\mathfrak{g}$ is semisimple if and only if $\mathfrak{z}(\mathfrak{g}) = 0$.

Definition 42 (radical). Fix a Lie algebra $\mathfrak{g}$. Then the *radical* $\text{rad } \mathfrak{g}$ is the sum of all solvable ideals of $\mathfrak{g}$. One can check that $\text{rad } \mathfrak{g}$ is a solvable ideal.

Example 43. For a Lie algebra $\mathfrak{g}$, we see $\text{rad } \mathfrak{g} = 0$ if and only if $\mathfrak{g}$ is semisimple. For example, $\mathfrak{g}/\text{rad } \mathfrak{g}$ is semisimple.

Theorem 44 (Levi decomposition). Fix any Lie algebra $\mathfrak{g}$. Then there is a semisimple Lie subalgebra $\mathfrak{g}_{\text{ss}} \subseteq \mathfrak{g}$ and a direct sum decomposition

$$\mathfrak{g} = \text{rad } \mathfrak{g} \oplus \mathfrak{g}_{\text{ss}}.$$

Sketch. This is a very difficult result. The idea is to show that the short exact sequence

$$0 \rightarrow \text{rad } \mathfrak{g} \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}_{\text{ss}} \rightarrow 0$$

admits a splitting. Such an extension problem reduces to a calculation of some cohomology groups, which is tricky. ■

Proposition 45. Let V be a representation of a Lie algebra $\mathfrak{g}$. Suppose that the pairing $\langle -, - \rangle_V : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ defined by

$$\langle X, Y \rangle_V := \text{tr}(XY; V)$$

is non-degenerate. Then $\mathfrak{g}$ is reductive.

Sketch. It is enough to show that any $X \in [\mathfrak{g}, \text{rad } \mathfrak{g}]$ lives in the kernel of $\langle -, - \rangle_V$ (and thus vanishes). By passing to some short exact sequences, we may assume that V is irreducible. Using Theorem 29, one can show that each $\text{rad}(\mathfrak{g})$ -eigenspace of V is $\mathfrak{g}$ -invariant, so irreducibility implies that $\text{rad } \mathfrak{g}$ acts by scalars on V . A direct calculation now shows that $[\mathfrak{g}, \text{rad } \mathfrak{g}]$ acts by zero on V . ■

Example 46. The standard representation $V = \mathbb{C}^n$ of $\mathfrak{gl}_n(\mathbb{C})$ satisfies the hypothesis of Proposition 45, so $\mathfrak{gl}_n(\mathbb{C})$ is reductive. A similar argument holds for $\mathfrak{so}_n(\mathbb{C})$.

Theorem 47 (Cartan). Let $\mathfrak{g}$ be a semisimple Lie algebra. Then the pairing

$$\langle X, Y \rangle_{\mathfrak{g}} := \operatorname{tr}(XY; \mathfrak{g})$$

is non-degenerate. This pairing is called the *Killing form*.

Sketch. Please skip this sketch. Let $\mathfrak{h}$ be the kernel of $\langle -, - \rangle_{\mathfrak{g}}$, which is an ideal. We will show that $\mathfrak{h}$ is a solvable, which forces $\mathfrak{h} = 0$. It is enough to show that $\operatorname{ad}(\mathfrak{h}) \subseteq \mathfrak{gl}(\mathfrak{h})$ is solvable. Some linear algebra shows that the pairing $\langle X, Y \rangle := \operatorname{tr}(XY; \mathfrak{h})$ on $\mathfrak{h}$ vanishes. Thus, we are in the situation that we have a Lie algebra $\mathfrak{h}'$ with a faithful representation V for which

$$\operatorname{tr}(XY; V) = 0$$

for each $X, Y \in \mathfrak{h}'$. One can use the Jordan decomposition to show that any element of $[\mathfrak{h}', \mathfrak{h}']$ is nilpotent, so $[\mathfrak{h}', \mathfrak{h}']$ is nilpotent, so $\mathfrak{h}'$ is solvable. ■

1.4 The Root Decomposition

Definition 48. Fix a Lie algebra $\mathfrak{g}$. Then $X \in \mathfrak{g}$ is *nilpotent* or *semisimple* if and only if $\operatorname{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ is nilpotent or semisimple, respectively.

Proposition 49 (Jordan decomposition). Fix a semisimple Lie algebra $\mathfrak{g}$. For any $X \in \mathfrak{g}$, there is a unique semisimple $X_s \in \mathfrak{g}$ and $X_n \in \mathfrak{g}$ such that $[X_s, X_n] = 0$ and $X = X_s + X_n$.

Sketch. The usual Jordan decomposition produces a unique pair $(X'_s, X'_n) \in \mathfrak{gl}(\mathfrak{g})^2$ for which X'_s is semisimple, X'_n is nilpotent, $[X'_s, X'_n] = 0$, and $\operatorname{ad}_X = X'_s + X'_n$. Thus, it remains to check that X'_s and X'_n are in the image of $\operatorname{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$. In short, the Jordan decomposition shows that X'_s and X'_n are polynomials in ad_X , so they are derivations of $\mathfrak{g}$, but all derivations of $\mathfrak{g}$ come from $\mathfrak{g}$. ■

Definition 50 (Cartan). Fix a semisimple Lie algebra $\mathfrak{g}$. Then a *torus* is an abelian subalgebra $\mathfrak{h}$ of $\mathfrak{g}$ consisting of semisimple elements. A maximal torus is a *Cartan subalgebra*.

Example 51. The subalgebra of diagonal matrices of $\mathfrak{sl}_n(\mathbb{C})$ is a Cartan subalgebra.

Remark 52. Let $\mathfrak{h}$ be a Cartan subalgebra. It turns out that

$$\mathfrak{h} = \{X \in \mathfrak{h} : [X, \mathfrak{h}] = 0\}.$$

In short, use the Jordan decomposition and the maximality of $\mathfrak{h}$.

Remark 53. It turns out that any two Cartan subalgebras of $\mathfrak{g}$ are conjugate under $\operatorname{Aut} \mathfrak{g}$. This is fairly technical, and we will only use it to show that certain constructions are independent of the choice of $\mathfrak{h}$.

Definition 54 (rank). Fix a semisimple Lie algebra $\mathfrak{g}$. Then the *rank* of $\mathfrak{g}$ is the dimension of a Cartan subalgebra of $\mathfrak{g}$.

Example 55. The rank of $\mathfrak{sl}_n(\mathbb{C})$ is $n - 1$.

Definition 56. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Then the *root system* $\Phi \subseteq \mathfrak{h}^*$ is the set of nonzero eigenspaces of $\mathfrak{h}$ acting on $\mathfrak{g}$.

Proposition 57. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Then the *root decomposition* is the eigenspace decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_{\alpha},$$

where $\mathfrak{g}_{\alpha}$ is the eigenspace of $\alpha \in \Phi$.

Sketch. Diagonalize the representation $\mathfrak{g}$ of $\mathfrak{h}$. ■

Remark 58. For $\alpha, \beta \in \Phi$, we have $[\mathfrak{g}_{\alpha}, \mathfrak{g}_{\beta}] \subseteq \mathfrak{g}_{\alpha+\beta}$.

Remark 59. The subspace Φ spans $\mathfrak{h}^*$. Indeed, any $h \in \mathfrak{h}$ orthogonal to all $\alpha \in \Phi$ then has $\text{ad}_h = 0$, so $h \in \mathfrak{z}(\mathfrak{g})$, so $h = 0$.

Notation 60. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$ and root system $\Phi \subseteq \mathfrak{h}^*$. For any $\alpha \in \Phi$, the subspaces $\mathfrak{g}_{\alpha}$ and $\mathfrak{g}_{-\alpha}$ generate a subalgebra of $\mathfrak{g}$ isomorphic to $\mathfrak{sl}_2(\mathbb{C})$. (This is a little tricky to show.) We denote this subalgebra by $\mathfrak{sl}_2(\mathbb{C})_{\alpha}$.

Proposition 61. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$ and root system $\Phi \subseteq \mathfrak{h}^*$. Let $(-, -)$ be the Killing form on $\mathfrak{g}$.

- (a) For each $\alpha \in \Phi$, $\dim \mathfrak{g}_{\alpha} = 1$.
- (b) For each $\alpha, \beta \in \Phi$, $2(\alpha, \beta)/(\alpha, \alpha) \in \mathbb{Z}$.
- (c) For each $\alpha \in \Phi$, define the reflection $s_{\alpha} : \mathfrak{h}^* \rightarrow \mathfrak{h}^*$ defined by $s_{\alpha}(\lambda) := \lambda - 2(\alpha, \lambda)/(\alpha, \alpha) \cdot \alpha$. Then s_{α} preserves Φ .
- (d) If $\alpha \in \Phi$ and $k\alpha \in \Phi$ for some $k \in \mathbb{C}$, then $k \in \{+1, -1\}$.

Sketch. Let $(-, -)$ be the Killing form on $\mathfrak{h}^*$, which is non-degenerate. Define $h_{\alpha} := 2H_{\alpha}/(\alpha, \alpha)$, where H_{α} is the element dual to $\alpha \in \mathfrak{h}^*$. Then one can show that

$$\mathbb{C}h_{\alpha} \oplus \bigoplus_{k \in \mathbb{Z} \setminus \{0\}} \mathfrak{g}_{k\alpha}$$

is an irreducible representation of $\mathfrak{sl}_2(\mathbb{C})_{\alpha}$. Each part now follows from understanding the irreducible representations of $\mathfrak{sl}_2(\mathbb{C})$. ■

Example 62. Set $\mathfrak{g} := \mathfrak{sl}_n(\mathbb{C})$, and let $\mathfrak{h}$ be the diagonal Cartan subalgebra. For each $i \in \{1, 2, \dots, n\}$, let $e_i : \mathfrak{h} \rightarrow \mathbb{C}$ be the i th projection. For each distinct i and j , the elementary matrix E_{ij} is an eigenvector for the adjoint $\mathfrak{h}$ -action with eigenvalue $e_i - e_j$. Thus, $\Phi = \{e_i - e_j : i \neq j\}$. Technically, $\mathfrak{h}^*$ should be thought of as $\mathbb{C}^n / \mathbb{C}(e_1 + \dots + e_n)$.

Remark 63. One can recover $\mathfrak{g}$ from the root system $\Phi \subseteq \mathfrak{h}^*$. Indeed, $\mathfrak{g}$ has basis given by a basis of $\mathfrak{h}^*$ and a vector e_α for each $\alpha \in \Phi$. Roughly speaking, the Lie bracket is determined by requiring $[e_\alpha, e_\beta] \in \mathbb{C}e_{\alpha+\beta}$ via the $\mathfrak{sl}_2(\mathbb{C})_\alpha$ -representation generated by e_β . A more precise version of this remark is given by the Serre presentation of a semisimple Lie algebra, which we will not discuss further.

1.5 Root Systems

Definition 64 (reflection). A Euclidean vector space E is a finite-dimensional real vector space equipped with an inner product $\langle -, - \rangle$. Fix $\alpha \in E$. Then define the reflection $s_\alpha \in O(E)$ as

$$s_\alpha(\lambda) := \lambda - \frac{2(\alpha, \lambda)}{(\alpha, \alpha)} \cdot \alpha$$

Definition 65 (root system). Fix a Euclidean vector space E . A root system is a nonempty finite subset $\Phi \subseteq E \setminus \{0\}$ such that

- (a) Φ spans E ;
- (b) for each $\alpha, \beta \in \Phi$, the number $n_{\alpha\beta} := 2(\alpha, \beta)/(\beta, \beta)$ is an integer; and
- (c) Φ is stable under s_α for each $\alpha \in \Phi$.

The root system Φ is *reduced* if and only if $\mathbb{Z}\alpha \cap \Phi = \{\pm\alpha\}$ for each $\alpha \in \Phi$. The *rank* of a root system is $\dim_{\mathbb{R}} E$.

Remark 66. By Cauchy-Schwarz, $n_{\alpha\beta} \in \{-2, -1, -1/2, 0, +1/2, +1, +2\}$.

Example 67. For any semisimple Lie algebra $\mathfrak{g}$ with Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Then the root system $\Phi \subseteq \mathfrak{h}^*$ is a reduced root system. The root system of $\mathfrak{sl}_n(\mathbb{C})$ is denoted A_{n-1} .

Example 68. If $\Phi_1 \subseteq E_1$ and $\Phi_2 \subseteq E_2$ are root systems, then $\Phi_1 \sqcup \Phi_2 \subseteq E_1 \times E_2$ is also a root system. Here, $\Phi_1 \subseteq E_1 \times E_2$ refers to the subset $\{(\alpha, 0) \in E_1 \times E_2 : \alpha \in \Phi_1\}$.

Definition 69 (coroot). Fix a Euclidean vector space E . For $\alpha \in E$, its coroot $\alpha^\vee \in E^*$ is defined by

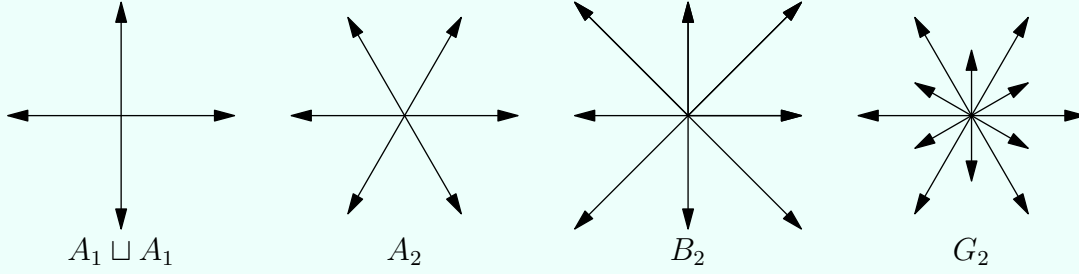
$$\alpha^\vee(\lambda) := \frac{2(\alpha, \lambda)}{(\alpha, \alpha)}.$$

Definition 70. Fix root systems $\Phi_1 \subseteq E_1$ and $\Phi_2 \subseteq E_2$. An *isomorphism* of root systems is a bijection $\varphi: \Phi_1 \rightarrow \Phi_2$ which extends to a linear map $E_1 \rightarrow E_2$ such that

$$\frac{2(\alpha, \beta)}{(\beta, \beta)} = \frac{2(\varphi(\alpha), \varphi(\beta))}{(\varphi(\beta), \varphi(\beta))}.$$

Remark 71. The linear map $\varphi: E_1 \rightarrow E_2$ does not have to preserve the inner product. For example, one could simply define E_2 to have twice the inner product of E_1 (and the same underlying vector space).

Example 72. Here are all the root systems of rank two, up to isomorphism, drawn in $\mathbb{R}^2$.



Definition 73 (irreducible). A root system Φ is *irreducible* if and only if it is not isomorphic to a root system of the form $\Phi_1 \sqcup \Phi_2$, where Φ_1 and Φ_2 are root systems.

Remark 74. It turns out that any root system is the disjoint union of irreducible root systems.

Definition 75 (Weyl group). Fix a root system $\Phi \subseteq E$. Then the *Weyl group* W of Φ is the subgroup of $GL(E)$ generated by $\{s_\alpha : \alpha \in \Phi\}$. The *sign* of $w \in W$, denoted $\det w$, is equal to the determinant of its action on E .

Example 76. The Weyl group of A_n is S_{n+1} .

1.6 Dynkin Diagrams

Definition 77 (polarization). Fix a root system $\Phi \subseteq E$. A *polarization* of Φ is a choice of $t \in E$ such that $(t, \alpha) \neq 0$ for all $\alpha \in \Phi$. The pair (Φ, t) is a *polarized root system*, though we will frequently omit the choice of t .

Definition 78 (positive). Fix a polarized root system $\Phi \subseteq E$. Then a root $\alpha \in \Phi$ is *positive* if and only if $(t, \alpha) > 0$. The set of positive roots is denoted Φ^+ . One similarly defines the set of negative roots Φ^- .

Example 79. The element $t := (1, 2, \dots, n)$ is a polarization of A_n . Then the positive roots are $\{e_i - e_j : i < j\}$.

Definition 80 (simple). Fix a polarized root system $\Phi \subseteq E$. Then a positive root $\alpha \in \Phi$ is *simple* if and only if it is not the sum of more than one positive root.

Remark 81. It turns out that the set of simple roots is a basis of E . Also, any positive root can be written as a sum of simple roots with nonnegative integral coefficients.

Example 82. The simple roots of A_n are $\{e_i - e_{i+1} : 1 \leq i < n\}$.

Lemma 83. Fix a root system $\Phi \subseteq E$. Then one can recover Φ from the subset $\Delta \subseteq \Phi$ of simple roots.

Sketch. Let W be the Weyl group of Φ . It turns out that the set $\{s_\alpha : \alpha \in \Delta\}$ of “simple reflections” generates W , and $W(\Delta) = \Phi$. ■

Definition 84 (Dynkin diagram). Fix a polarized root system $\Phi \subseteq E$, and enumerate the simple roots as $\{\alpha_1, \dots, \alpha_n\}$. Then the *Dynkin diagram* of Φ is the directed multigraph with the following data.

- The vertices of Φ is the index set $\{1, \dots, n\}$.
- Vertices i and j are connected by $a_{ij}a_{ji}$, with the edge directed from i to j if and only if $(\alpha_i, \alpha_i) > (\alpha_j, \alpha_j)$.

Remark 85. The Dynkin diagram D of a root system Φ is disconnected if and only if Φ is reducible, and each connected component of D corresponds to an irreducible component of Φ .

Example 86. The Dynkin diagram of A_n is the path with n vertices.

Lemma 87. Fix a root system $\Phi \subseteq E$. Then one can recover Φ from its Dynkin diagram.

Sketch. It is enough to recover the simple roots. Well, the Dynkin diagram remembers the number of simple roots and the numbers $n_{ij} := 2(\alpha_i, \alpha_j)/(\alpha_j, \alpha_j)$, which is enough to recover the root system up to isomorphism. ■

Theorem 88. The set of Dynkin diagrams of all possible irreducible root systems are as follows. (A graph labeled $?_n$ is understood to have n vertices.)

Name	Graph
A_n	• — • — ... — • — •
B_n	• — • — ... — • — •
C_n	• — • — ... — • $\Leftarrow$ •
D_n	• — • — ... — • $\swarrow$ • $\searrow$ •
E_4, E_5, E_6	• — • — • — ... — • $\uparrow$ •
F_4	• — • $\Rightarrow$ • — •
G_2	• $\Rightarrow$ •

Sketch. There are two parts to this proof. First, one has to construct a root system giving each of these root systems. Second, one has to show that all Dynkin diagrams have this form. There is nothing too interesting about the first step. For the second step, the idea is to constrain the numbers $2(\alpha_i, \alpha_j)/(\alpha_i, \alpha_i)$ that can appear in a root system. For example, because $(-, -)$ is positive-definite, the matrix $[(\alpha_i, \alpha_j)]_{ij}$ is positive-definite. One can use this to rule out certain subgraphs from appearing in the Dynkin diagram, such as cycles. ■

Remark 89. By Remark 63 and Lemma 87, this classifies all simple complex Lie algebras.

1.7 Highest Weight Theory

Convention 90. Throughout this section, $\mathfrak{g}$ is a semisimple Lie algebra with a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. The polarized root system is $\Phi \subseteq \mathfrak{h}^*$, and its Weyl group is W . Representations need not be finite-dimensional.

Theorem 91. All finite-dimensional representations of $\mathfrak{g}$ are completely reducible.

Sketch. There are many proofs of this theorem. The main idea is to show that any short exact sequence

$$0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0$$

of representations of $\mathfrak{g}$ splits as a direct sum $V = V' \oplus V''$. The most algebraic proof proceeds by finding “Casimir” elements in the center of $U\mathfrak{g}$. By Lemma 11, they act by scalars on irreducible representations. One can force the splitting if granted a Casimir elements which acts on V' and V'' by different scalars. This turns out to be possible as long as V' and V'' are not isomorphic. The case where V' and V'' are isomorphic is handled directly. ■

Definition 92 (weight decomposition). A *weight* is an element of $\mathfrak{h}^*$. For a representation V of $\mathfrak{g}$, a *weight vector* is an eigenvector for $\mathfrak{h}$, and a *weight decomposition* is an eigenspace decomposition of $\mathfrak{h}$ acting on V . For $\lambda \in \mathfrak{h}^*$, we define $V[\lambda]$ to be the λ -eigenspace, meaning

$$V[\lambda] := \{v \in V : Xv = \lambda(X)v \text{ for all } X \in \mathfrak{h}\}.$$

The multiset of $\lambda \in \mathfrak{h}^*$ for which $V[\lambda] \neq 0$ are referred to as the *weights* of V , where multiplicities are by $\dim_{\mathbb{C}} V[\lambda]$.

Definition 93 (integral). A weight $\lambda \in \mathfrak{h}^*$ is *integral* if and only if $\alpha^\vee(\lambda) \in \mathbb{Z}$ for all $\alpha \in \Phi$. The subgroup of integral weights is denoted $P \subseteq \mathfrak{h}^*$. While we’re here, we define the *root lattice* $Q \subseteq \mathfrak{h}^*$.

Remark 94. The definition of the root system implies that $Q \subseteq P$. The quotient P/Q is interesting.

Example 95. For $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{C})$, one has $P = \{(\lambda_1, \dots, \lambda_n) : \lambda_i - \lambda_j \in \mathbb{Z} \text{ for all } i, j\}$ and $Q = \{(\lambda_1, \dots, \lambda_n) : \sum_i \lambda_i = 0\}$. Thus, $P/Q \cong \mathbb{Z}/n\mathbb{Z}$.

Lemma 96. Any finite-dimensional representation V of $\mathfrak{g}$ admits a weight decomposition by integral weights. The multiset of weights of V is W -invariant.

Sketch. Let’s start with the weight decomposition. Because $\mathfrak{h}$ is abelian and spanned by $\{h_\alpha\}_{\alpha \in \Phi}$, it is enough to show that h_α acts semisimply on V for each $\alpha \in \Phi$. For a given $\alpha \in \Phi$, this follows by the representation theory of $\mathfrak{sl}_2(\mathbb{C})$ upon restricting to $\mathfrak{sl}_2(\mathbb{C})_\alpha \subseteq \mathfrak{g}$. Integrality follows because h_α acts on $V[\lambda]$ by $\alpha^\vee(\lambda)$, which must be an integer. To check invariance under W , it is enough to check invariance under each reflection s_α for $\alpha \in \Phi$, which follows by the symmetry of the weights of the irreducible representations of $\mathfrak{sl}_2(\mathbb{C})_\alpha$. ■

Notation 97. Define $\mathfrak{n}_+$ to be

$$\bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha \subseteq \mathfrak{g}.$$

Define $\mathfrak{n}_-$ similarly. We may abbreviate $\mathfrak{n}_+$ to $\mathfrak{n}$. These are Lie subalgebras of $\mathfrak{g}$.

Definition 98 (singular). Fix a representation V of $\mathfrak{g}$. A vector $v \in V$ is *singular* if and only if v is a nonzero weight vector and $\mathfrak{n} \cdot v = 0$. The representation V is a *highest weight* representation of weight λ if and only if V is generated by a singular vector of weight λ .

Lemma 99. Fix an irreducible finite-dimensional representation V of $\mathfrak{g}$. Then V is a highest weight representation.

Sketch. By irreducibility, V is generated by any nonzero vector, so it is enough to exhibit a nonzero singular vector. Select the weight λ in V which is “maximal” in the sense that $\alpha + \lambda$ is not a weight in V for any $\alpha \in \Phi^+$. Then any nonzero $v \in V[\lambda]$ will do. ■

Definition 100 (Verma module). Fix a weight $\lambda \in \mathfrak{h}^*$. Then define the *Verma module* M_λ to be $U\mathfrak{g} \otimes_{\mathfrak{b}} \mathbb{C}_\lambda$, where $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}$. For brevity, we define $v_\lambda \in M_\lambda$ to be the vector $1 \otimes 1$.

Remark 101. By Theorem 15, the natural map $U\mathfrak{n}_- \rightarrow U\mathfrak{g}$ induces an isomorphism $U\mathfrak{n}_- \rightarrow M_\lambda$ of vector spaces, meaning that M_λ admits a basis by ordered monomials in $\mathfrak{n}_-$.

Lemma 102. Fix a representation V of highest weight λ . For any singular vector $v \in V[\lambda]$, there is a unique surjection $M_\lambda \rightarrow V$ such that $v_\lambda \mapsto v$.

Sketch. There is a natural map $U\mathfrak{g} \rightarrow V$ sending $1 \mapsto v$. Because v is a highest weight vector of weight λ , this natural map is surjective and descends to M_λ . ■

Lemma 103. Fix $\lambda \in \mathfrak{h}^*$. Then M_λ admits a unique maximal proper submodule.

Sketch. A submodule N of M_λ is proper if and only if $v_\lambda \notin N$, so the maximal submodule is simply the sum of all submodules not containing v_λ . ■

Notation 104. Fix a weight $\lambda \in \mathfrak{h}^*$. Then define L_λ to the quotient of M_λ by its maximal proper submodule.

Remark 105. Assembling our results shows that any finite-dimensional irreducible representation V is of highest weight λ for some weight λ and thus must be isomorphic to L_λ .

Definition 106 (dominant). A weight $\lambda \in \mathfrak{h}^*$ is *dominant* if and only if $\alpha^\vee(\lambda) \geq 0$ for each $\alpha \in \Phi^+$.

Theorem 107 (Highest weight). For $\lambda \in \mathfrak{h}^*$, the representation L_λ is finite-dimensional if and only if λ is dominant and integral. Thus, finite-dimensional irreducible representations are in bijection with dominant integral weights.

Sketch. If L_λ is finite-dimensional, then for each $\alpha \in \Phi^+$, one knows that v_λ generates a finite-dimensional representation of $\mathfrak{sl}_2(\mathbb{C})_\alpha$ with $e_\alpha v_\lambda = 0$ and $h_\alpha v_\lambda = \alpha^\vee(\lambda) v_\lambda$. Comparing this with the known representations of $\mathfrak{sl}_2(\mathbb{C})$ forces $\alpha^\vee(\lambda)$ to be a nonnegative integer.

Conversely, suppose λ is dominant and integral, so we must check $\dim_{\mathbb{C}} L_\lambda < \infty$. This is long and difficult, so we will say very little. In short, for each simple reflection s , one can show that $M_{s\lambda}$ embeds into M_λ , and then one should check that the quotient of M by the sum of the $M_{s\lambda}$ s is finite-dimensional. ■

Remark 108. Though we have classified the finite-dimensional irreducible representations, this construction is highly non-explicit. It is possible but difficult to even get rough information about these representations; see Theorem 118 and Corollary 120.

1.8 Character Theory

Convention 109. Convention 90 continues to apply.

Definition 110 (character). Fix a representation V of $\mathfrak{g}$. Then the *character* of V is the formal power series

$$\text{ch } V := \sum_{\lambda \in \mathfrak{h}^*} \dim_{\mathbb{C}} V[\lambda] \cdot e^\lambda.$$

Here, e^λ is a formal symbol satisfying the usual exponent rule $e^{\lambda+\mu} = e^\lambda e^\mu$ for $\lambda, \mu \in \mathfrak{h}^*$.

Example 111. Let $\mathbb{C}$ be the trivial representation. Then $\text{ch } \mathbb{C} = 1$.

Example 112. Let $\mathbb{C}^2$ be the standard representation of $\mathfrak{sl}_2(\mathbb{C})$. Then $\text{ch } \mathbb{C}^2 = e^{-\alpha/2} + e^{\alpha/2}$, where $\alpha \in \mathfrak{h}$ is the unique positive root. More generally,

$$\text{ch } \text{Sym } \mathbb{C}^2 = e^{-n\alpha/2} + e^{-n\alpha/2+1} + \dots + e^{n\alpha/2-1} + e^{n\alpha/2}.$$

Example 113. For any representation V , one has $\text{ch } V^\vee = \overline{\text{ch } V}$, where the overline means $\overline{e^\lambda} = e^{-\lambda}$.

Example 114. For representations V and W , one has $\text{ch}(V \oplus W) = \text{ch } V + \text{ch } W$. More generally, given a short exact sequence

$$0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0$$

of representations admitting weight decompositions, one has $\text{ch } V = \text{ch } V' + \text{ch } V''$.

Example 115. For representations V and W , one has $\text{ch}(V \otimes W) = \text{ch } V \cdot \text{ch } W$.

Example 116. Fix $\lambda \in \mathfrak{h}^*$. By expanding out the basis of M_λ , one finds that

$$\text{ch } M_\lambda = e^\lambda \prod_{\alpha \in \Phi^+} \frac{1}{1 - e^{-\alpha}}.$$

Notation 117. Define $\rho := \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha$.

Theorem 118 (Weyl character formula). Fix a dominant integral weight λ . Then

$$\text{ch } L_\lambda = \sum_{w \in W} \det(w) e^{w(\lambda + \rho)} \Bigg/ \prod_{\alpha \in \Phi^+} (e^{\alpha/2} - e^{-\alpha/2}).$$

Sketch. This is another long and difficult theorem. One approach is to build an exact sequence of the form

$$0 \rightarrow ? \rightarrow ? \rightarrow \cdots \rightarrow ? \rightarrow L_\lambda \rightarrow 0,$$

where each $?$ is a sum of Verma modules. For example, there is a natural surjection $M_\lambda \rightarrow L_\lambda$, and the kernel admits a natural surjection of the form $\bigoplus_i M_{s_i \lambda}$. Once one fills in each $?$, one can apply Example 116. ■

Example 119 (Weyl demoninator formula). When $\lambda = 0$, we find $L_0 = \mathbb{C}$, so $\text{ch } L_0 = 1$. Thus,

$$\sum_{w \in W} \det(w) e^{w\rho} = \prod_{\alpha \in \Phi^+} (e^{\alpha/2} - e^{-\alpha/2}).$$

Corollary 120 (Weyl dimension formula). Fix a dominant integral weight λ . Then

$$\dim L_\lambda = \prod_{\alpha \in \Phi^+} \frac{(\lambda + \rho, \alpha)}{(\rho, \alpha)}.$$

Sketch. Fix a formal variable q , and we will compute

$$\dim_q L_\lambda := \sum_{\mu \in \mathfrak{h}^*} \dim_{\mathbb{C}} L_\lambda[\mu] \cdot q^{2(\rho, \lambda)},$$

from which the dimension can be computed by sending $q \rightarrow 1$. There is a homomorphism $\mathbb{C}[e^P] \rightarrow \mathbb{C}[q, 1/q]$ given by sending e^λ to $q^{2(\lambda, \rho)}$, and it sends $\text{ch } L_\lambda$ to $\dim_q L_\lambda$. One now computes $\dim_q L_\lambda$ by using Theorem 118: note

$$\dim_q L_\lambda = \sum_{w \in W} \det(w) q^{(w(\lambda + \rho), \rho)} \Bigg/ \prod_{\alpha \in \Phi^+} (q^{(\rho, \alpha)} - q^{-(\rho, \alpha)}).$$

The numerator can be simplified with Example 119, and then we may take $q \rightarrow 1$. ■

A Type A_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{sl}(n+1)$.
- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_{n+1}) : a_1 + \dots + a_{n+1} = 0\}$.
- $\mathfrak{h}^*$ is spanned by the projections e_i : $\text{diag}(a_1, \dots, a_{n+1}) \mapsto a_i$, where $e_1 + \dots + e_{n+1} = 0$.
- Killing form $K(X, Y) = 2(n+1) \text{tr } XY$.

- Root system

root	root space	coroot
$e_i - e_j$	E_{ij}	$E_{ii} - E_{jj}$

- Lattices

- Weight lattice $P = \{(\lambda_1, \dots, \lambda_{n+1}) \in \mathbb{R}^{n+1} / (1, \dots, 1)\mathbb{R} : \lambda_i - \lambda_{i+1} \in \mathbb{Z} \text{ for all } i\}$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n\}$.
- Root lattice $Q = \{(\lambda_1, \dots, \lambda_{n+1}) \in \mathbb{Z}^{n+1} : \lambda_1 + \dots + \lambda_{n+1} = 0\}$.
- $P/Q \cong \mathbb{Z}/(n+1)\mathbb{Z}$ by $(\lambda_1, \dots, \lambda_{n+1}) \mapsto \sum_i \lambda_i$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\}$.
- The simple roots are $\{e_i - e_{i+1} : i\}$. Set $\alpha_i := e_i - e_{i+1}$.
- Dynkin diagram:

$$\alpha_1 \text{ --- } \alpha_2 \text{ --- } \dots \text{ --- } \alpha_n$$

- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones.
- Maximal root $\theta = e_{n+1} - e_n$.

- Weyl group

- Weyl group is $W = S_{n+1}$ acting by permutations of coordinates.
- The simple reflection s_i is $(i, i+1)$.
- $w_0 = (n+1, n, \dots, 2, 1)$.

- Representation theory

- $\rho = (n/2, n/2 - 1, \dots, 1 - n/2, -n/2)$ and $\rho^\vee = (n/2, n/2 - 1, \dots, 1 - n/2, -n/2)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_i : i\}$.
- Fundamental representations are $L_{\omega_i} = \wedge^i \mathbb{C}^{n+1}$.
- Exponents are $\{1, 2, \dots, n\}$.

- Group theory

- $G_{\text{sc}} = \text{SL}_{n+1}(\mathbb{C})$.
- $G_{\text{ad}} = \text{PSL}_{n+1}(\mathbb{C})$.

B Type B_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{so}(2n+1)$, preserving

$$B = \begin{bmatrix} & & 1_n \\ & 1_n & \\ & & \\ & & & 1 \end{bmatrix}.$$

- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n, 0) : a_1, \dots, a_n\}$.
- $\mathfrak{h}^*$ is spanned by the projections $e_i : \text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n, 0) \mapsto a_i$.

- Root system

root	length	root space	coroot
$e_i - e_j$	long	$E_{ij} - E_{i+n, j+n}$	$H_i - H_j$
$e_i + e_j$	long	$E_{i, j+n} - E_{j, i+n}$	$H_i + H_j$
$-e_i - e_j$	long	$E_{i+n, j} - E_{j+n, i}$	$-H_i - H_j$
e_i	short	$E_{i, 2n+1} - E_{2n+1, i+n}$	$2H_i$
$-e_i$	short	$E_{n+i, 2n+1} - E_{2n+1, i}$	$-2H_i$

Here, $H_i = E_{ii} - E_{i+n, i+n}$.

- Lattices

- Weight lattice $P = \{(\lambda_1, \dots, \lambda_n) \in (\frac{1}{2}\mathbb{Z})^n : \lambda_i - \lambda_{i+1} \in \mathbb{Z} \text{ for all } i\}$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n \geq 0\}$.
- Root lattice $Q = \mathbb{Z}^n$.
- $P/Q \cong \mathbb{Z}/2\mathbb{Z}$ by $(\lambda_1, \dots, \lambda_n) \mapsto \lambda_1$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\} \cup \{e_i + e_j : i < j\} \cup \{e_i : i\}$.
- The simple roots are $\{e_i - e_{i+1} : i\} \cup \{e_n\}$. Set $\alpha_i := e_i - e_{i+1}$ and $\alpha_n := e_n$.
- Dynkin diagram:

$$\alpha_1 \text{ --- } \alpha_2 \text{ --- } \dots \text{ --- } \alpha_{n-1} \implies \alpha_n$$

- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones for $i < n$ and $\omega_n := \frac{1}{2}(1, \dots, 1)$.
- Maximal root $\theta = e_1 + e_2$.

- Combinatorics

- Weyl group is $W = S_n \ltimes \{\pm 1\}^n$ acting by permutations of coordinates and sign changes.
- The simple reflection s_i is $(i, i+1)$ for $i < n$, and s_n is $(+1, \dots, +1, -1)$.
- $w_0 = (-1, \dots, -1)$.

- Representation theory

- $\rho = (n - 1/2, n - 3/2, \dots, 3/2, 1/2)$ and $\rho^\vee = (n, n - 1, \dots, 2, 1)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_1\}$.
- Fundamental representations are $L_{\omega_i} = \wedge^i \mathbb{C}^{2n+1}$ for $i < n$ and the standard Clifford module for $i = n$.
- Exponents are $\{1, 3, \dots, 2n - 3, 2n - 1\}$.

- Group theory

- $G_{\text{sc}} = \text{Spin}_{2n+1}(\mathbb{C})$.
- $G_{\text{ad}} = \text{SO}_{2n+1}(\mathbb{C})$.

C Type C_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{sp}(2n)$, preserving

$$B = \begin{bmatrix} & -1_n \\ 1_n & \end{bmatrix}.$$

- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) : a_1, \dots, a_n\}$.
- $\mathfrak{h}^*$ is spanned by the projections $e_i : \text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) \mapsto a_i$.

- Root system

root	length	root space	coroot
$e_i - e_j$	short	$E_{ij} - E_{i+n, j+n}$	$H_i - H_j$
$e_i + e_j$	short	$E_{i, j+n} - E_{j, i+n}$	$H_i + H_j$
$-e_i - e_j$	short	$E_{i+n, j} - E_{j+n, i}$	$-H_i - H_j$
$2e_i$	long	$E_{i, i+n}$	H_i
$-2e_i$	long	$E_{i+n, i}$	$-H_i$

Here, $H_i = E_{ii} - E_{i+n, i+n}$.

- Lattices

- Weight lattice $P = \mathbb{Z}^n$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n \geq 0\}$.
- Root lattice $Q = \{(\lambda_1, \dots, \lambda_n) \in \mathbb{Z}^n : \lambda_1 + \dots + \lambda_n \equiv 0 \pmod{2}\}$.
- $P/Q \cong \mathbb{Z}/2\mathbb{Z}$ by $(\lambda_1, \dots, \lambda_n) \mapsto \sum_i \lambda_i$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\} \cup \{e_i + e_j : i < j\} \cup \{2e_i : i\}$.
- The simple roots are $\{e_i - e_{i+1} : i\} \cup \{2e_n\}$. Set $\alpha_i := e_i - e_{i+1}$ and $\alpha_n := 2e_n$.
- Dynkin diagram:

$$\alpha_1 \text{ --- } \alpha_2 \text{ --- } \dots \text{ --- } \alpha_{n-1} \text{ } \Longleftarrow \alpha_n$$

- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones.
- Maximal root $\theta = 2e_1$.

- Combinatorics

- Weyl group is $W = S_n \ltimes \{\pm 1\}^n$ acting by permutations of coordinates and sign changes.
- The simple reflection s_i is $(i, i+1)$ for $i < n$, and s_n is $(+1, \dots, +1, -1)$.
- $w_0 = (-1, \dots, -1)$.

- Representation theory

- $\rho = (n, n-1, \dots, 2, 1)$ and $\rho^\vee = (n-1/2, n-3/2, \dots, 3/2, 1/2)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_1\}$.
- Fundamental representations are $L_{\omega_i} = \ker(\iota_B : \wedge^i \mathbb{C}^{2n} \rightarrow \wedge^{i-2} \mathbb{C}^{2n})$, where ι_B is contraction:

$$\iota_B(v_1 \wedge \dots \wedge v_i) := \sum_{r < s} (-1)^{r+s+1} B(v_r \wedge v_s)(v_1 \wedge \dots \wedge \widehat{v_r} \wedge \dots \wedge \widehat{v_s} \wedge \dots \wedge v_i).$$

- Exponents are $\{1, 3, \dots, 2n-3, 2n-1\}$.

- Group theory

- $G_{\text{sc}} = \text{Sp}_{2n}(\mathbb{C})$.
- $G_{\text{ad}} = \text{Sp}_{2n}(\mathbb{C})/\{\pm 1\}$.

D Type D_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{so}(2n)$, preserving

$$B = \begin{bmatrix} & 1_n \\ 1_n & \end{bmatrix}.$$

- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) : a_1, \dots, a_n\}$.
- $\mathfrak{h}^*$ is spanned by the projections $e_i : \text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) \mapsto a_i$.

- Root system

root	root space	coroot
$e_i - e_j$	$E_{ij} - E_{j+n, i+n}$	$H_i - H_j$
$e_i + e_j$	$E_{i, j+n} - E_{j, i+n}$	$H_i + H_j$
$-e_i - e_j$	$E_{i+n, j} - E_{j+n, i}$	$-H_i - H_j$

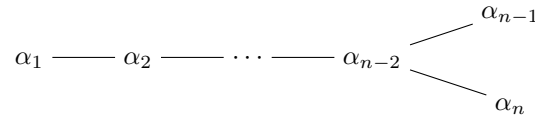
Here, $H_i = E_{ii} - E_{i+n, i+n}$.

- Lattices

- Weight lattice $P = \{(\lambda_1, \dots, \lambda_n) \in (\frac{1}{2}\mathbb{Z})^n : \lambda_i - \lambda_{i+1} \in \mathbb{Z} \text{ for all } i\}$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n, \lambda_{n-1} + \lambda_n \geq 0\}$.
- Root lattice $P = \{(\lambda_1, \dots, \lambda_n) \in \mathbb{Z}^n : \lambda_1 + \dots + \lambda_n \equiv 0 \pmod{2}\}$.
- $P/Q \cong (\mathbb{Z}/2\mathbb{Z})^2$ if n even and $P/Q \cong \mathbb{Z}/4\mathbb{Z}$ if n odd. It is generated by $(1, 0, \dots, 0)$ and $\frac{1}{2}(1, \dots, 1)$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\} \cup \{e_i + e_j : i < j\}$.
- The simple roots are $\{e_i - e_{i+1} : i\} \cup \{e_{n-1} + e_n\}$. Set $\alpha_i := e_i - e_{i+1}$ and $\alpha_n := e_{n-1} + e_n$.
- Dynkin diagram:



- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones for $i < n-1$ and $\omega_{n-1} := \frac{1}{2}(1, \dots, 1, -1)$ and $\omega_n := \frac{1}{2}(1, \dots, 1)$.
- Maximal root $\theta = e_1 + e_2$.

- Combinatorics

- Weyl group is $W = S_n \ltimes \{\pm 1\}_+^n$ acting by permutations of coordinates and even sign changes.
- The simple reflection s_i is $(i, i+1)$ for $i < n$, and s_n is $(+1, \dots, +1, -1, -1)$.
- If n even, $w_0 = (-1, \dots, -1)$ acts by (-1) on P/Q . If n odd, $w_0 = (-1, \dots, -1, +1)$ does not.

- Representation theory

- $\rho = (n-1, n-2, \dots, 1, 0)$ and $\rho^\vee = (n-1, n-2, \dots, 1, 0)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_1, \omega_{n-1}, \omega_n\}$.
- Fundamental representations are $L_{\omega_i} = \wedge^i \mathbb{C}^{2n+1}$ for $i < n$ and the degree pieces of the standard Clifford module.
- Exponents are $\{1, 3, \dots, 2n-3, 2n-1\} \sqcup \{n\}$.

- Group theory

- $G_{\text{sc}} = \text{Spin}_{2n}(\mathbb{C})$.
- $G_{\text{ad}} = \text{SO}_{2n}(\mathbb{C})/\{\pm 1\}$.